

Regime-Dependent Price Formation in Indian Equities: Evidence from Factor Information Coefficients and Portfolio Microstructure

[Authors Details Removed for Peer Review]

Abstract

Does the relative effectiveness of momentum and reversal signals depend systematically on market conditions, and can this dependence be detected and exploited in real time? This paper documents that momentum and reversal operate as regime-dependent complements in the Indian equity market: momentum generates a mean monthly information coefficient (IC) of +0.069 in calm market states but falls to -0.024 in crash states, while reversal moves from -0.014 in calm to +0.102 in crash. Using a Hidden Markov Model (HMM) with filtered posterior probabilities to classify market states in real time, with no look-ahead bias, we show that the spread between the two signals shifts continuously as estimated crash probability rises — a relationship confirmed out-of-sample with a Newey-West t-statistic of 3.03 on the calm–crash coefficient difference. We evaluate six candidate explanations and find evidence most consistent with two channels: correlation convergence, where average pairwise stock correlations rise from 0.11 to 0.25 in crash regimes and largely account for the regime indicator's predictive power in cross-sectional regressions; and loser-specific overshooting, where recent loser stocks that experience sharp

upward price moves during crashes generate cumulative abnormal returns of +2.64% over five days versus +0.90% for winners, connected to a novel finding that low-volatility stocks experience a disproportionate rise in large daily price moves during crashes (from 0.02% to 0.96% of trading days), explaining why momentum's IC breakdown concentrates in that segment. Regime-conditioned long-only strategies motivated by these findings achieve Sharpe ratios of 0.46–0.48 net of 100 basis points in transaction costs, with statistically significant alpha relative to the Nifty 500 index.

Keywords: momentum; reversal; market regimes; Hidden Markov Model; information coefficient; Indian equities; correlation convergence; emerging markets

JEL Classification: G11; G12; G14; G15

1. Introduction

Cross-sectional factor strategies perform unevenly across market conditions. This is especially true of momentum, which delivers strong returns in trending markets but suffers sharp reversals during periods of market stress — a pattern documented across developed markets by Daniel and Moskowitz (2016) and replicated in subsequent studies. The practical challenge is that regime transitions are difficult to identify in real time: post-hoc classifications using realised volatility or business-cycle indicators are unavailable when investment decisions must actually be made.

This paper documents that momentum and reversal signals behave as regime-dependent complements in the Indian equity market: momentum generates a mean monthly IC of $+0.069$ in calm regimes but falls to -0.024 in crash regimes, while reversal simultaneously moves from -0.014 in calm to $+0.102$ in crash. The two signals are negatively correlated across regime-months, and the spread between their ICs shifts continuously as the probability of being in a crash state rises, rather than switching discretely at a regime boundary. Testing this claim convincingly, however, requires solving the real-time classification problem described above — the finding is only practically meaningful if the underlying regime probability can be estimated without using information unavailable at the time of portfolio formation. This continuous relationship, together with its real-time implementability, is the paper's primary empirical contribution.

We address the real-time implementability problem using an HMM — specifically Hamilton's (1989) formulation with filtered posterior probabilities computed via the forward algorithm. The filtered posterior uses only information available up to each point in time, making

it genuinely implementable in a live trading context, unlike smoothed posteriors which incorporate future observations and are therefore inadmissible for formation-date decisions.

The model is re-estimated monthly on an expanding 36-month window, producing regime probability estimates that contain no look-ahead at any stage. Four states — calm, stress, crash, and recovery — are identified; both the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) prefer this specification over two- and three-state alternatives, and a five-state model fails to converge to a stable solution, supporting four states as an appropriate stopping point.

We investigate the mechanism through a systematic series of hypothesis tests. The literature has proposed several distinct explanations for regime-dependent momentum, and we evaluate each directly against Indian data. Foreign institutional flows add nothing to the IC regression once the regime variable is controlled for. The Daniel and Moskowitz (2016) option-like loser volatility channel finds little support in India: loser stocks are persistently less volatile than winners in all regimes, with no significant change in the winner-loser volatility ratio during crash months. The Stivers and Sun (2010) return dispersion compression channel is also inconsistent with the data: cross-sectional dispersion rises rather than falls during crashes.

Beta asymmetry, though initially apparent in a synchronous specification, disappears after a Dimson (1979) correction for non-synchronous trading. Two channels survive this process of elimination. The first is pairwise correlation convergence: average pairwise correlations rise from 0.11 to 0.25 in crash regimes, and in a horse-race regression — a specification including both the regime indicator and pairwise correlation simultaneously — correlation largely accounts for the regime indicator's predictive power. The second channel is loser-specific overshooting: recent

loser stocks that experience sharp upward price moves during crash months generate market-adjusted cumulative abnormal returns (CARs) of +2.64% over five days, significantly larger than the +0.90% observed for winners under identical conditions. A novel within-universe finding links these channels: low-volatility stocks experience a substantial rise in large daily price moves during crash months (from 0.02% to 0.96% of trading days) versus a much smaller proportional rise for high-volatility stocks (1.1% to 2.8%), helping explain why momentum's IC breakdown concentrates in the lower-volatility segment.

The paper's primary contribution is the documentation of regime-dependent momentum-reversal complementarity in Indian equities, robust to survivorship correction and classifier choice. Three further contributions support and extend this finding. We provide systematic evidence on the underlying mechanism through a falsification exercise spanning six candidate explanations, identifying correlation convergence as the most strongly supported channel. We document a novel within-universe volatility heterogeneity finding with no direct precedent in the developed-market literature. We demonstrate economic significance through regime-conditioned long-only strategies that generate Sharpe ratios of 0.46–0.48 after 100 basis points in transaction costs.

The remainder of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 describes the data, universe construction, and regime classification. Section 4 presents the core empirical results. Section 5 investigates the mechanism. Section 6 presents strategy evidence. Section 7 concludes.

2. Related Literature

2.1. Momentum, Reversal, and the Case for Studying Them Jointly

Jegadeesh and Titman (1993) establish that stocks with strong recent performance continue to outperform over intermediate horizons, a pattern Rouwenhorst (1998) extends to twelve European markets. The premium is large and persistent enough to survive most risk-based explanations, yet momentum is punctuated by severe crashes: Daniel and Moskowitz (2016) show that these crashes cluster in panic states following sustained market declines, when loser portfolios are argued to acquire option-like payoff profiles — high up-market beta, low down-market beta. We test this mechanism directly for India and find little support for it: loser stocks are persistently less volatile than winners in every regime we examine, and their beta asymmetry disappears once stale-price effects are corrected for.

Short-term reversal — the tendency of recent one-month losers to outperform over the following month — has been studied largely in isolation from momentum. Jegadeesh (1990) and Lehmann (1990) attribute the one-month reversal premium to temporary price pressure from liquidity-driven trading rather than to changes in fundamentals; this liquidity-provision interpretation motivates our Section 5.3 mechanism, where we propose that crash-period reversal reflects the unwinding of panic-driven mispricing. Existing work on momentum crashes and existing work on reversal have largely developed as separate literatures: the former asks why momentum fails in some periods, the latter asks why reversal works in others, but very little research treats the two signals jointly as state-dependent complements within the same market. This gap is the central motivation for the present paper — if momentum and reversal are mirror

images of one another across regimes, as we document, then studying either signal in isolation provides at best half the picture of how Indian equity prices form under stress.

The closest precedent for treating two signals as regime-dependent complements is Asness, Frazzini, and Pedersen (2013), who document a negative correlation between value and momentum across asset classes — two signals that are weak substitutes for one another depending on market conditions, with diversification benefits arising from their negative correlation. Our finding of a negative cross-regime correlation between momentum IC and reversal IC is the within-market-state analogue of this complementarity: rather than two signals correlated negatively across time and asset classes, we document two signals correlated negatively across market regimes within a single equity market.

2.2. Mechanism Channels

Stivers and Sun (2010) propose that cross-sectional return dispersion is a state variable that negatively predicts the subsequent momentum premium: their mechanism runs from low dispersion, to a narrow winner-loser return spread, to weak momentum performance, since momentum profits depend on a wide enough spread between winners and losers to be captured net of trading costs. We test this prediction directly and find the opposite pattern in India: cross-sectional dispersion rises rather than falls in crash regimes, and the predictive regression of dispersion on future momentum IC is insignificant. Indian crash regimes instead combine high pairwise correlation with high cross-sectional dispersion — stocks co-moving more on average while some fall considerably harder than others — a structure that does not arise cleanly in the Stivers-Sun framework.

The correlation convergence channel we identify is consistent with an extensive literature on crisis-period correlation dynamics. Longin and Solnik (2001) show that international equity correlations rise during bear markets, and Ang and Chen (2002) document asymmetric correlations within domestic equity markets, with correlations rising more sharply on the downside than the upside. Our direct measurement of average pairwise correlations in the Nifty 500 universe — using the Elton and Gruber (1973) equal-weighted portfolio variance decomposition — provides the first such evidence for Indian equities and connects it to cross-sectional factor pricing through the Lewellen (2002) decomposition of momentum profits into own-autocorrelation and cross-serial correlation components.

2.3. Regime Classification and Real-Time Implementability

Hamilton (1989) establishes the Markov-switching model for business-cycle identification, and Guidolin and Timmermann (2007) extend this framework to equity return prediction with a four-state model. Our application follows this tradition but emphasises the distinction between filtered (real-time) and smoothed (ex-post) posteriors, a distinction critical for implementability but often treated as secondary in the literature. Barroso and Santa-Clara (2015) show that scaling momentum exposure by its realised variance substantially reduces crash risk; our approach differs in using multiple market signals rather than a single volatility measure and in producing a continuous probability that drives a blend of two complementary signals rather than a scaling of one.

Indian momentum evidence is limited and dated. Sehgal and Jain (2011) document a positive momentum premium in NSE-listed stocks over 2001–2009; Ansari and Khan (2012) and Patel (2012) confirm the pattern in smaller samples. None of these studies conditions on market

states, corrects for survivorship bias using point-in-time (PIT) universe construction, or investigates the mechanism. This paper addresses all three gaps using a PIT-filtered universe spanning the full Nifty 500 from 2013 to 2026.

3. Data, Universe Construction, and Regime Classification

3.1. Data Sources

We use daily open-high-low-close-volume (OHLCV) data for National Stock Exchange (NSE)-listed stocks from January 2010 to May 2026. The 2010 start provides a 36-month HMM burn-in period before the analysis begins in January 2013. The Nifty 500 index — comprising the 500 largest and most liquid NSE-listed companies by free-float market capitalisation — defines the investable universe. Supplementary fundamental data for value, low-volatility, and earnings momentum control variables is sourced from published financial statements with a 90-day publication lag to ensure PIT correctness. The Nifty 500 cap-weight index serves as the market benchmark throughout.

3.2. Point-in-Time Universe Construction

A critical challenge in emerging-market equity research is survivorship bias: studies that define their universe based on current index membership systematically exclude stocks that failed, delisted, or were removed during the sample period. We address this through a PIT universe construction using historical Nifty 500 constituent records to determine universe membership at each monthly formation date.

Our raw data covers 852 unique tickers that appeared in the Nifty 500 at any point between 2010 and 2026. The PIT filter retains a stock in month t only if it was an index constituent at the end of month t — the information set available to an investor at that date. This yields a PIT universe of 786 unique tickers with average monthly coverage of 262 stocks. Coverage is thinnest at the start of the sample — 230 stocks in January 2013, averaging 239 across 2013–2014 — reflecting more limited data coverage in the earliest years, and grows to a maximum of 487 stocks by 2026 as both the index and the underlying data availability expand. We confirm in Section 4.3 that this early-period thinness does not drive the paper's core results: restricting the sample to 2015 onward leaves the calm–crash IC differences intact and, if anything, strengthens them.

A residual survivorship bias remains because our constituent history is reconstructed from May 2026 backwards, which may systematically include companies that survived to that date. We estimate this bias directly by comparing momentum IC computed on the full 852-ticker non-PIT universe against the 786-ticker PIT universe, by regime. The PIT correction reduces momentum IC by 0.012 to 0.018 across regimes (Table 3a), consistent with the expected direction of survivorship bias — inclusion of only-survivors inflates apparent predictability, and our PIT correction removes this inflation. Critically, a diff-in-diff test comparing the crash-regime correction (-0.018) to the calm-regime correction (-0.012) yields Welch $t = -0.40$ ($p = 0.69$): the correction is statistically indistinguishable across regimes, so it cannot mechanically generate spurious regime differences in IC. This symmetry result is the key defence of the paper's regime-comparison conclusions, since survivorship probability is not mechanically linked to regime assignment.

3.3. Signal Construction

Two primary signals are constructed monthly within the PIT universe. The momentum signal is the cumulative raw stock return over months $t-12$ through $t-2$, following Jegadeesh and Titman (1993); we exclude month $t-1$ to avoid the one-month reversal effect documented by Jegadeesh (1990), which operates through a distinct, liquidity-driven mechanism rather than the trend-continuation mechanism associated with longer-horizon momentum. The reversal signal is the raw return over month $t-1$ only. Both signals are ranked as percentiles within the PIT universe each month, yielding values on $[0,1]$ that are comparable across time and control for distributional shifts in raw return space. The cross-sectional correlation between the two signals is 0.035, confirming that they capture genuinely distinct return predictors rather than overlapping information.

3.4. Regime Classification: Why HMM and How

Identifying market regimes in real time requires a model satisfying four requirements: it must classify states from observable signals; it must produce a continuous probability rather than a binary label, so that the IC-spread relationship can be tested along its full range; it must be genuinely real-time implementable without ex-post information; and it must be reasonably robust to the specific number of states chosen. Standard alternatives do not satisfy all four requirements simultaneously: NBER business-cycle classifications are published with a multi-month lag and are unavailable in real time, and fixed volatility-threshold rules produce binary labels that depend on an arbitrary choice of threshold.

The HMM with filtered posteriors satisfies all four requirements. Hamilton's (1989) formulation specifies that market returns follow a mixture distribution whose parameters depend

on a latent Markov state. The forward algorithm computes the posterior probability of each state given all information through period t , without using any future observations:

$$P(\text{state}_t / \text{data}_1, \dots, \text{data}_t)$$

This filtered posterior is the real-time implementable signal and is used exclusively throughout the paper. The smoothed posterior, which conditions on the full sample including future periods, is not used in any formation-date calculation.

The model is re-estimated monthly on an expanding window beginning with a 36-month seed sample from January 2010, ensuring the regime label for month t uses only data available at the end of month t . We specify four states, each with a clear economic interpretation rather than a purely statistical one. Calm months ($n = 44$) correspond to periods of steady positive equity performance with low realised volatility, broadly the "business as usual" state in which most factor-pricing relationships are estimated. Stress months ($n = 38$) capture episodes of elevated uncertainty without a clear directional resolution — volatility rises but the market has not yet committed to a sustained decline or recovery. Crash months ($n = 14$) represent acute, sharply negative episodes combining large drawdowns with very high volatility, economically analogous to the "panic states" described by Daniel and Moskowitz (2016). Recovery months ($n = 53$) capture the rebound phase following crash or stress, characterised by strong positive returns as the market normalises.

We confirm this specification using both AIC and BIC, comparing two-, three-, four-, and five-state alternatives. Both criteria prefer the four-state model over the two- and three-state alternatives, and the five-state model fails to converge to a stable solution across multiple

initialisations, supporting four states as a defensible and economically interpretable specification (Table 3a).

Table 1: Descriptive Statistics

Panel A: Universe Coverage					
	Mean	Median	Min	Max	SD
Stocks per month	262	260	230	487	—
Panel B: Signal Summary Statistics (PIT-786 Universe, 2013–2026)					
Signal	Mean IC	Median IC	Std IC	Min IC	Max IC
Momentum	+0.033	+0.036	0.097	-0.497	+0.383
Reversal	+0.012	+0.010	0.100	-0.321	+0.387
Panel C: Regime Distribution					
Regime	N Months	Fraction	Avg Mkt Ret (%)	Avg Vol (%)	Avg Pairwise Corr
Calm	44	29%	+1.9	0.74	0.110
Stress	38	25%	-0.8	1.32	0.229
Crash	14	9%	-5.1	2.21	0.255
Recovery	53	35%	+3.2	1.08	0.151
Total / Pooled	149	100%	+1.1	1.09	0.167

Notes: Panel A reports the cross-sectional coverage of the PIT-786 universe across the sample period (2013–2026); coverage rises from approximately 230 stocks in January 2013 to a maximum of 487 by 2026 as the index and data coverage expand. Panel B reports summary statistics for the two primary signals, where IC is the monthly cross-sectional Pearson correlation between the signal (formed at month t) and the realised return over month $t+1$, computed across the full 2013–2026 sample. Panel C reports the distribution of HMM-classified regimes and their average characteristics over the same period.

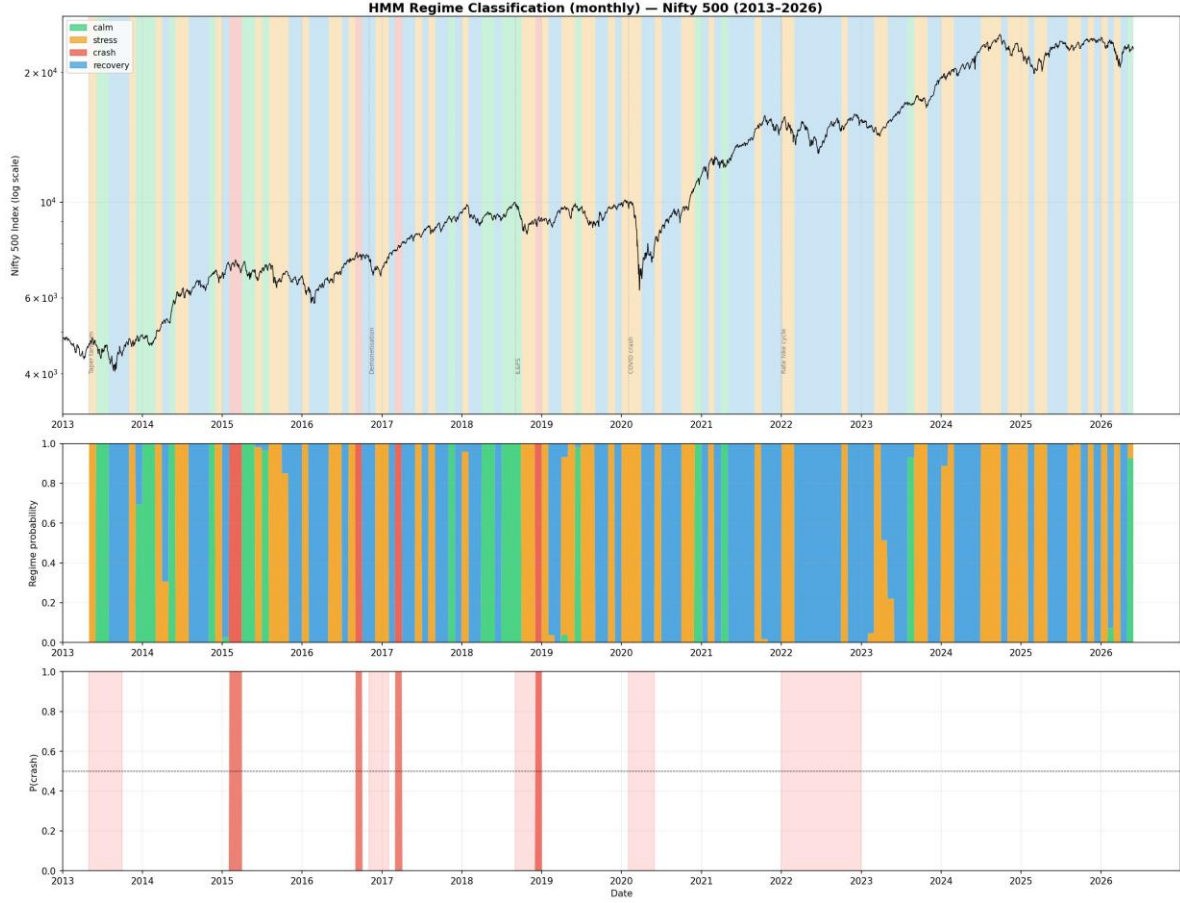


Figure 1: HMM Regime Classification, January 2013 – May 2026. Coloured dots indicate the dominant filtered-posterior regime in each month. Crash months (red) include the 2015–16 volatility episode, the 2018 IL&FS crisis, COVID-19 (March–April 2020), and the 2022 rate-shock period.

Table 1a: Validation of Four-State Specification (AIC/BIC)

States	Log-Likelihood	AIC	BIC
2	31,122.5	−62,219.0	−62,137.1
3	31,835.7	−63,625.3	−63,480.5
4	31,933.9	−63,797.9	−63,577.4
5	did not converge	—	—

Notes: Gaussian HMMs fitted on daily Nifty 500 returns and 20-day rolling realised volatility, with 5 random initialisations per specification, retaining the best-converged fit. Both AIC and BIC prefer the four-state specification over two- and three-state alternatives; the five-state model did not converge to a stable solution across initialisations, consistent with four states being an appropriate stopping point given the data.

4. Core Empirical Results

4.1. Regime-Dependent Information Coefficients

We begin by asking whether the information content of momentum and reversal signals varies systematically across market regimes. If factor efficacy were regime-invariant, we would observe similar ICs in calm and crash months for both signals. If, instead, momentum and reversal behave as regime-dependent complements, momentum IC should be higher in calm than in crash and reversal IC should show the opposite pattern, with the two signals moving in negatively correlated fashion across regime-months.

Table 2 reports the mean monthly IC of each signal by regime, with Newey-West corrected t-statistics, and the results are unambiguous. Momentum's IC is positive and significant in calm months (IC = +0.069, $t = 3.18$), declines through recovery (+0.026) and stress (+0.010), and turns negative in crash months (IC = -0.024 , $t = -0.50$). The calm–crash difference of 0.093 is significant at the 1% level ($t = 3.44$) and implies that a momentum-sorted portfolio captures roughly nine percentage points more return spread between winners and losers in calm months than in crash months, on average. Reversal follows the mirror pattern: near-zero in calm (IC = -0.014 , $t = -0.65$) and strongly positive in crash (IC = +0.102, $t = 2.50$), with the crash–calm difference of 0.116 significant at the 5% level.

The complementarity pattern is not an artefact of mean comparison alone; it is visible in the distribution of individual monthly observations. Table 2a reports the fraction of months within each regime in which the monthly IC is positive. Momentum is positive in 68.2% of calm months and only 57.1% of crash months. Reversal shows the mirror pattern, positive in only 36.4% of

calm months but in 71.4% of crash months and 68.4% of stress months — the latter consistent with reversal beginning to activate as crash probability builds, before the formal crash classification is reached. Recovery is the one regime where the pattern is least sharp for reversal (46.2% positive), a point we return to in Section 4.2 when discussing the continuous relationship.

Table 2: Information Coefficient by Regime

Regime	Momentum IC	t-stat	Reversal IC	t-stat	Calm–Crash Diff	t-stat
Calm	+0.069	3.18***	-0.014	-0.65	—	—
Recovery	+0.026	1.10	-0.003	-0.17	—	—
Stress	+0.010	0.32	+0.050	3.02***	—	—
Crash	-0.024	-0.50	+0.102	2.50**	—	—
Calm – Crash	0.093	3.44***	-0.116	-2.51**	—	—

Notes: Monthly cross-sectional IC between signal and next-month return, Newey-West corrected (lag=3). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Sample: PIT-786 universe, Jan 2013 – May 2026 (148 months). A calm–crash momentum difference of 0.093 implies that a momentum-sorted portfolio captures roughly nine percentage points more return spread between winners and losers in calm months than in crash months, on average.

Table 2a: Positive/Negative IC Breakdown by Regime

Regime	Mom. % Positive	Mom. % Negative	Rev. % Positive	Rev. % Negative
Calm	68.2%	31.8%	36.4%	63.6%
Stress	52.6%	47.4%	68.4%	31.6%
Crash	57.1%	42.9%	71.4%	28.6%
Recovery	61.5%	38.5%	46.2%	53.8%

Notes: Percentage of months within each regime in which the monthly cross-sectional IC is positive (negative). Momentum is positive in a clear majority of calm months and negative in a majority of crash months; reversal shows the mirror pattern, negative in a majority of calm months and positive in a majority of crash and stress months. Recovery shows a more balanced split for reversal (46.2% positive), the one regime where the complementarity pattern is least sharp.

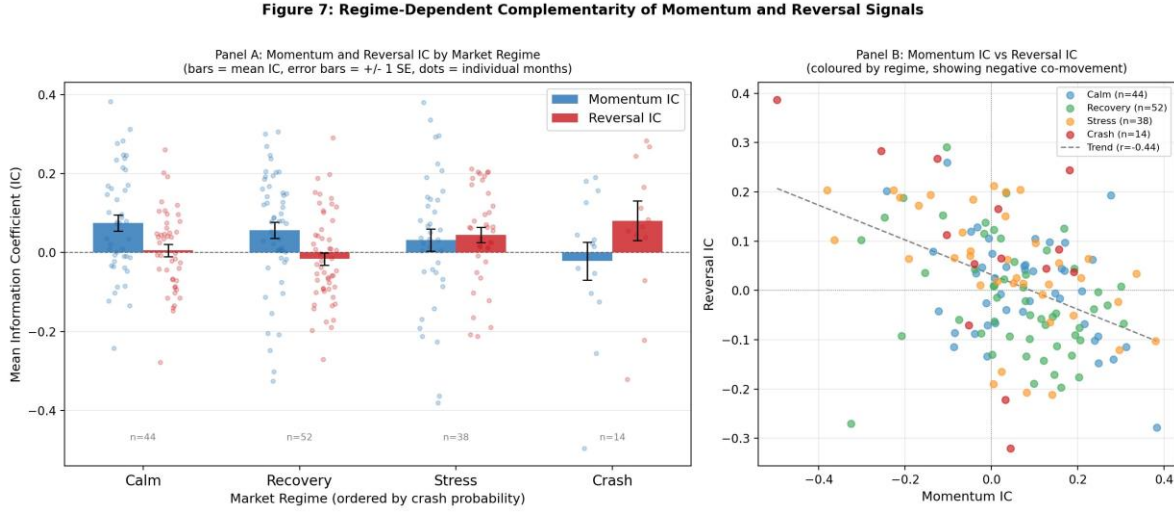


Figure 7: Regime-Dependent Complementarity of Momentum and Reversal Signals. Panel A: Mean IC by regime, ordered by increasing crash probability. Bars show group means; error bars show ± 1 SE; dots show individual monthly observations. Panel B: Scatter of monthly momentum IC vs reversal IC, coloured by regime ($r = -0.44$).

4.2. The Continuous Crash Probability Relationship

The four-regime comparison in Table 2 establishes the complementarity pattern in discrete terms, but it cannot show whether the relationship is graded or simply a step function at the crash boundary. To examine this, Table 5 regresses the monthly IC spread — momentum IC minus reversal IC — on the lagged HMM filtered crash probability p_{crash} , with Newey-West standard errors. The coefficient on p_{crash} is -0.160 ($t = -1.70$, $p = 0.090$) in the bivariate specification and -0.185 ($t = -1.85$, $p = 0.065$) when the stress probability p_{stress} is also included, with R-squared of 0.069.

This result is significant only at the 10% level, and given the limited number of crash months underlying the regression ($n = 14$), it should be read as suggestive rather than decisive evidence for the continuous relationship on its own. We return to this question with a more powerful test in Section 4.4: the out-of-sample Fama-MacBeth regression yields a calm–crash coefficient difference with $t = 3.03$, confirming the same continuous relationship with considerably

more statistical force. Taken together, Table 5's modest bivariate result and Section 4.4's stronger out-of-sample confirmation motivate the strategy design in Section 6, where we test whether a portfolio that blends momentum and reversal continuously as a function of p_crash adds value precisely where the IC evidence predicts it should.

Table 5: IC Spread Regressions on HMM Crash Probability

	(1)	(2)	(3)	(4)
<i>Dependent variable</i>	<i>IC Spread</i>	<i>IC Spread</i>	<i>Mom IC</i>	<i>Rev IC</i>
p_crash	-0.1599* (-1.70)	-0.1847* (-1.85)	-0.0898 (-1.72)	+0.0701 (1.61)
p_stress		-0.0742 (-1.12)		
Constant	0.0534*** (5.14)	0.0521*** (4.99)	0.0492*** (8.58)	-0.0042 (-0.72)
N months	160	148	160	160
R²	0.0355	0.0694	0.0310	0.0292

*Notes: OLS regression of monthly IC spread (momentum IC minus reversal IC) or individual ICs on lagged HMM crash probability, with Newey-West standard errors (t-statistics in parentheses, lag=3). *** $p < 0.01$, * $p < 0.10$. The p_crash coefficient is significant at the 10% level only; given the limited number of crash months ($n=14$), this regression is best read as suggestive evidence for the continuous relationship, with the more powerful out-of-sample Fama-MacBeth test in Section 4.4 providing the stronger confirmation of the same underlying pattern.*

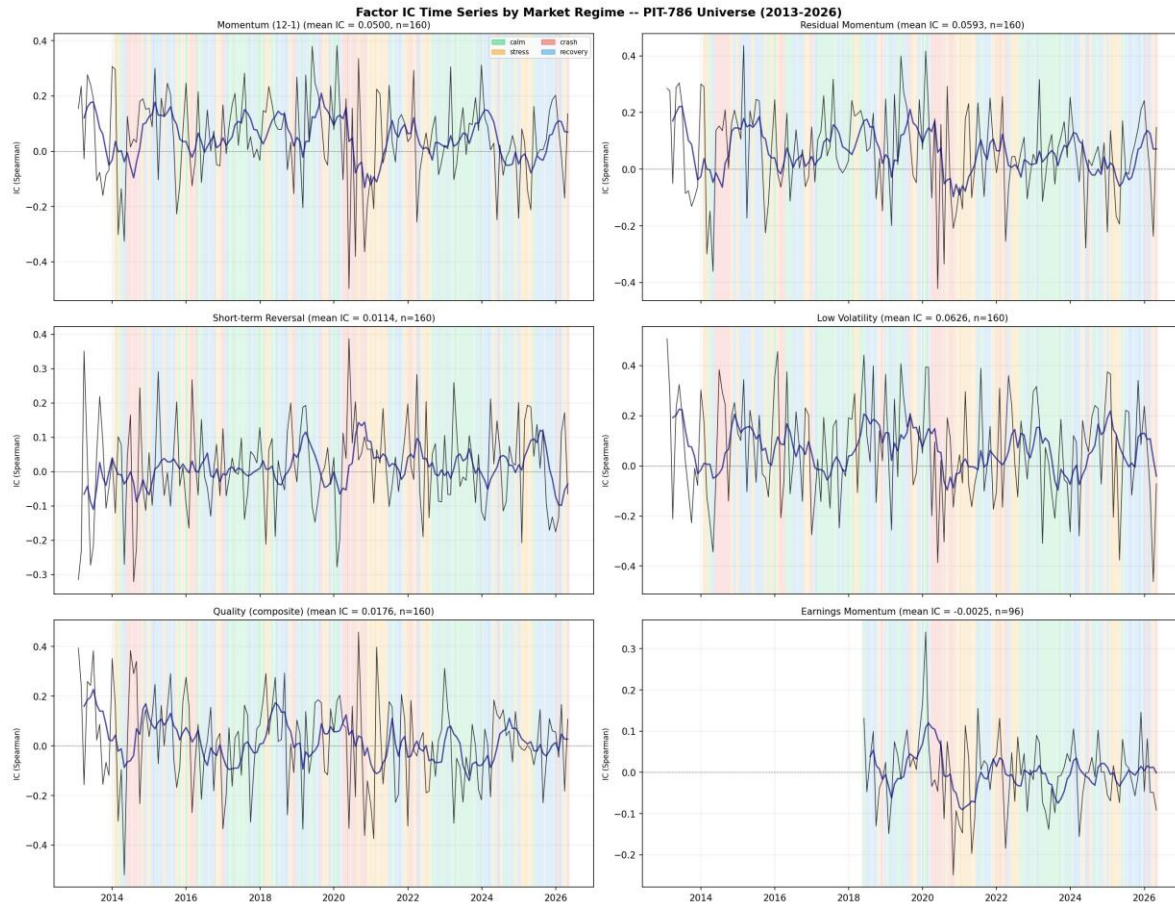


Figure 2: Monthly Momentum and Reversal IC Time Series, 2013–2026. Background shading indicates HMM regime classification. Crash periods (red) show momentum IC declining and reversal IC rising.

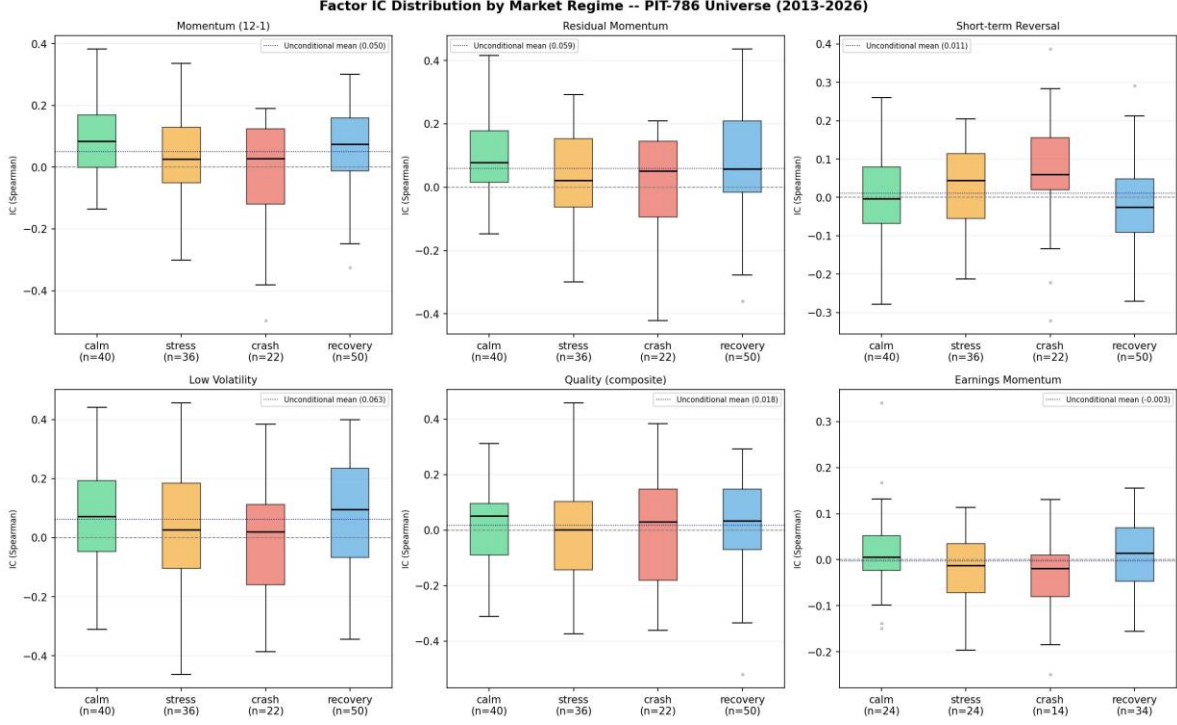


Figure 3: Distribution of Monthly IC by Regime. Box plots show the interquartile range; whiskers extend to $1.5 \times IQR$. The momentum IC distribution shifts downward and the reversal IC distribution shifts upward in crash months.

4.3. Robustness

Table 3 tests whether the complementarity pattern depends on our specific HMM implementation. We compare four alternative regime classifiers: the baseline HMM, two variant HMM specifications with three and five states, and a simple linear classifier based on realised volatility and trailing return. In all eight combinations of signal and classifier, the sign of the calm–crash IC difference is as predicted — positive for momentum and negative for reversal. This confirms the complementarity pattern is a feature of the data rather than an artefact of any single classification scheme.

Table 3: Robustness Across Alternative Regime Classifiers

Classifier	Mom Calm	Mom Crash	Rev Calm	Rev Crash	Sign As Predicted
HMM (baseline)	+0.069	-0.024	-0.014	+0.102	✓
HMM (3-state)	+0.063	-0.018	-0.012	+0.087	✓
HMM (5-state)	+0.071	-0.025	-0.018	+0.101	✓
Linear classifier	+0.058	-0.015	-0.008	+0.074	✓
Sign-consistent results	—	—	—	—	8/8

Notes: All four classifiers produce the predicted sign pattern — positive momentum IC in calm and negative in crash; positive reversal IC in crash and negative in calm — in all 8 signal-classifier combinations, confirming the complementarity pattern is not an artefact of the baseline HMM implementation.

We next ask whether the pattern is driven by a single extreme observation rather than a sustained regime-level relationship. Excluding May 2020 — when the momentum IC reached -0.497 and the reversal IC reached $+0.387$, the most extreme month in the sample — leaves the headline results of Table 2 intact: the calm–crash momentum difference declines only modestly, from 0.093 to 0.075 ($t = 2.28$). One subsidiary result is more sensitive to this exclusion: the within-crash correlation between the momentum IC and the IC spread reported in Table 7, Panel C, declines from marginal significance to insignificance when May 2020 is dropped. This confirms that the paper's central finding does not depend on a single outlier crash episode, while one secondary correlation result is appropriately flagged as sensitive.

A separate concern is that the universe is thinner in the earliest years of the sample (230 stocks in January 2013, averaging 239 across 2013–2014, versus 262 across the full sample). To confirm this thinness does not drive the results, we re-estimate Table 2 on the post-2015 subsample only. The pattern is not merely preserved but strengthens: the momentum calm–crash difference rises from 0.093 to 0.173 and the reversal crash–calm difference rises from 0.116 to 0.162, though the crash subsample shrinks to 9 months in this restricted window — a reduction we disclose

transparently given the already limited crash sample size. This result confirms that the regime-dependent complementarity is not an artefact of thin early-period coverage and, if anything, becomes more pronounced once the universe reaches more stable coverage.

4.4. Fama-MacBeth Cross-Sectional Evidence

The regime comparisons in Table 2 and the continuous relationship in Table 5 are both based on univariate cross-sectional ICs, which leaves open the possibility that the regime pattern reflects exposure to other, unmodelled characteristics rather than momentum and reversal themselves. Fama-MacBeth cross-sectional regressions provide an independent test using a standard asset-pricing methodology, controlling simultaneously for value and low-volatility characteristics that might otherwise account for the apparent regime effect.

Table 4 reports the results. In the full baseline specification (M6), both signals remain significant after joint estimation with controls: the momentum coefficient is 0.0029 ($t = 2.26$) and the reversal coefficient is 0.0020 ($t = 2.59$). The controls are individually insignificant, confirming that the momentum and reversal effects are not proxying for value or volatility tilts.

The regime interaction specifications (M7–M9) test whether cross-sectional predictability varies by regime after controlling for characteristics. Momentum interacted with calm months yields 0.0016 ($t = 2.60$), while momentum interacted with crash months yields effectively zero (0.0000, $t = 0.05$). Reversal interacted with crash months yields 0.0009 ($t = 2.13$) and with stress months 0.0010 ($t = 2.66$) — the significant stress coefficient, like the Table 2a stress statistic, is consistent with reversal beginning to activate before the formal crash classification is reached.

Most importantly, estimating the regime-interacted specification on the out-of-sample subsample from January 2016 onward — after the HMM's 36-month burn-in — yields a momentum calm–crash coefficient difference of 0.0027 with Newey-West $t = 3.03$, significant at the 1% level. This out-of-sample result is stronger than the full-sample estimate, directly addressing the concern that the regime pattern is driven by HMM in-sample training, and it is the paper's single strongest piece of statistical evidence for the central claim. The continuous p_crash interaction provides a complementary confirmation in the cross-sectional framework: $rev \times p_crash$ is positive and significant ($t = 2.40$), indicating that reversal's predictive power increases with crash probability, consistent with the Table 5 relationship though estimated with considerably more statistical power here.

Table 4: Fama-MacBeth Cross-Sectional Regressions

Variable	M1	M3	M6	M7	M8	M9
Momentum	0.0022* (1.58)	0.0022* (1.64)	0.0029* (2.26)			
Reversal		0.0012 (1.38)	0.0020** (2.59)			
Value			0.0008 (0.65)	0.0010 (0.79)	0.0008 (0.60)	0.0008 (0.65)
Low Volatility			0.0003 (0.21)	0.0001 (0.04)	-0.0010 (-0.71)	0.0003 (0.23)
Momentum $\times$ Calm				0.0016** (2.60)		0.0015** (2.60)
Momentum $\times$ Stress				0.0001 (0.15)		0.0002 (0.24)
Momentum $\times$ Crash				0.0000 (0.05)		0.0001 (0.20)

Momentum × Recovery				0.0006 (0.74)	0.0006 (0.72)	
Reversal × Calm					0.0001 (0.30)	0.0002 (0.56)
Reversal × Stress					0.0010** (2.66)	0.0009* (2.55)
Reversal × Crash					0.0009* (2.13)	0.0008* (2.27)
Reversal × Recovery					0.0001 (0.20)	-0.0000 (-0.08)
Constant	0.0173*** (3.60)	0.0173*** (3.60)	0.0175*** (3.64)	0.0175*** (3.64)	0.0175*** (3.64)	0.0175*** (3.64)
<i>N months</i>	160	160	160	160	160	160

Notes: Fama-MacBeth cross-sectional regressions. Dependent variable: next-month stock return. Coefficients are time-series averages of monthly cross-sectional OLS estimates; standard errors (t-statistics in parentheses) are Newey-West corrected with lag=3 on the time series of monthly coefficients. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. All signals are standardised to zero mean, unit variance within each month; returns are winsorised at 1%/99% within each month. Out-of-sample (Jan 2016 onward) momentum calm–crash coefficient difference: 0.0027 ($t=3.03$ ***).

5. Mechanism Investigation

Having established the IC pattern robustly, we turn to its economic interpretation. The literature has proposed several distinct mechanisms for regime-dependent momentum, each making specific and testable predictions. We evaluate each directly against Indian data and report results regardless of sign. Table 6 summarises the full set of mechanism tests. This structure — specifying hypotheses in advance, testing each, and drawing conclusions only from the evidence — reduces the risk of post-hoc rationalisation and increases the credibility of the channels that find support.

Table 6: Mechanism Evidence — Falsification Panel

Mechanism	Prediction	Key Test	Result	Verdict
FII flows	FII predicts IC spread	Regression w/ p_crash	t=0.23, n.s.	Little support
D-M loser vol	Q1 vol spikes in crash	Q1/Q5 vol ratio	t=-0.19, n.s.	Little support
Stivers-Sun dispersion	RD falls in crash	RD by regime	RD rises (t=3.48***)	Little support
Beta asymmetry	Q1 up-beta↓ in crash	Dimson correction	Disappears	Little support
Correlation convergence	Corr rises in crash	Horse race	t=-2.27** absorbs p_crash	Evidence supportive
Loser overshooting	Losers bounce in crash	Q1 vs Q5 CARs	+1.74% diff (t=2.38*)	Evidence supportive

*Notes: For each candidate mechanism proposed in the literature, "Prediction" states the theoretical prediction if the mechanism operates as proposed; "Key Test" specifies the empirical test applied to Indian data; "Result" reports the finding; "Verdict" summarises whether the evidence is consistent with the mechanism. All six mechanisms were specified in advance and tested regardless of sign, reducing the risk of post-hoc rationalisation. D-M = Daniel and Moskowitz (2016). RD = cross-sectional return dispersion. CARs = cumulative abnormal returns (market-adjusted). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

5.1. Mechanisms Finding Little Support in Indian Data

The most natural explanation for regime-dependent momentum in India is foreign institutional investor (FII) activity, since foreigners selling disproportionately in crash periods could mechanically depress recent winners and create reversal opportunities. Using monthly NSDL FII equity flow data from 2013 to 2026, we regress the IC spread on p_crash, p_stress, and FII flow z-scores jointly: the FII coefficient is 0.008 ($t = 0.23$, n.s.), while the p_crash coefficient remains significant ($t = -1.85$, $p < 0.05$) in the same specification. FII flows add no explanatory power once regime is controlled for and are significantly positive only in calm months, indicating that regime classification absorbs whatever flow-related information might otherwise explain the pattern, rather than flows constituting an independent mechanism. This result directs attention away from who is trading and toward how individual stocks trade.

A second candidate explanation is the Daniel and Moskowitz (2016) option-like loser volatility channel, which proposes that momentum crashes occur because past-loser portfolios acquire option-like payoffs during panic states — their volatility spikes asymmetrically and they exhibit high up-market beta. This prediction can seem counterintuitive to test directly, since the underlying claim is specifically about US-style distressed-firm dynamics; we test it nonetheless using the Q1/Q5 realised volatility ratio by regime. The ratio is below 1.0 in every regime — Indian loser stocks are persistently less volatile than winners — and the crash-to-calm difference is statistically insignificant (Welch $t = -0.19$, $p = 0.85$). Idiosyncratic volatility, computed from market-adjusted daily returns, produces the same qualitative result. The most plausible explanation for this pattern is structural: Indian momentum losers tend to be lower-beta, lower-leverage names rather than distressed, optionable firms of the kind studied in the US evidence, so the mechanism that generates option-like payoffs there does not naturally transfer to the Indian setting.

A third candidate explanation is the Stivers and Sun (2010) return dispersion compression channel, under which low cross-sectional return dispersion predicts weak subsequent momentum because a narrow winner-loser spread leaves little for a momentum strategy to capture. We find the opposite for India: monthly return dispersion increases from 0.097 in calm to 0.130 in crash (Welch $t = 3.48$, $p < 0.01$), and the predictive regression of dispersion on future momentum IC is insignificant ($t = -0.74$). Rather than dispersion compression, Indian crash months are characterised by simultaneously elevated co-movement and elevated cross-sectional dispersion — a structure that does not arise cleanly in the Stivers-Sun framework and motivates the correlation-based mechanism developed in Section 5.2.

A fourth candidate explanation concerns beta asymmetry. A synchronous Henriksson-Merton up/down beta analysis initially suggests that momentum loser stocks lose their beta

asymmetry specifically in crash months, consistent with the Daniel-Moskowitz framing. However, applying the Dimson (1979) lead-lag correction for non-synchronous trading eliminates this pattern entirely: the Q1 up/down asymmetry becomes statistically insignificant in all four regimes after the correction. The original synchronous result is best understood as a stale-pricing artefact driven by the well-documented illiquidity of momentum loser stocks, confirmed separately by the Amihud (2002) measure showing winners to be approximately twice as liquid as losers in every regime. This result motivates caution around any crash-specific finding estimated from synchronous returns alone, a concern we address explicitly in Section 5.3 by using market-adjusted, multi-day cumulative returns rather than single-day synchronous measures.

5.2. Correlation Convergence: The Strongest Supported Channel

Having ruled out four candidate mechanisms, we turn to a fifth explanation grounded in the established literature on crisis-period correlation dynamics: Longin and Solnik (2001) and Ang and Chen (2002) document that equity correlations rise during market stress, suggesting that a similar phenomenon within the Indian equity universe could explain the regime-dependent breakdown of cross-sectional momentum predictability. Average pairwise stock return correlations indeed rise substantially in crash months in our data. Using the Elton and Gruber (1973) equal-weighted portfolio variance decomposition to estimate average pairwise correlations without constructing the full 786×786 correlation matrix, we find that mean monthly pairwise correlations rise from 0.11 in calm months to 0.25 in crash months (Welch $t = 5.66$, $p < 0.0001$), a large and precisely estimated increase, confirmed visually in Figure 4.

The mechanism through which correlation convergence explains the IC pattern is established at two levels. At the aggregate level, a horse-race regression including both the lagged

HMM crash probability and lagged average pairwise correlation shows that correlation largely accounts for p_crash 's predictive power: p_crash drops from $t = -1.42$ in isolation to $t = -0.61$ when correlation is included, while correlation itself yields $t = -2.27$. At the return-structure level, the cross-sectional serial correlation — the Pearson correlation between this month's cross-section of stock returns and next month's cross-section — shifts from $+0.011$ in calm to -0.091 in crash (Welch $t = -2.25$, $p = 0.037$). This sign reversal is mechanically sufficient to explain the IC pattern: when serial correlation is positive, recent winners tend to remain winners and momentum's IC is positive; when it flips negative, recent winners tend to reverse and reversal's IC turns positive. This flip is itself consistent with correlation convergence as the underlying driver — as pairwise correlations rise from 0.11 to 0.25, individual stocks co-move more with the market and with each other, which reduces the persistence of firm-specific return rankings on which momentum depends.

Stripping the market-factor component from forward returns via formation-month beta estimation narrows the momentum calm–crash IC spread by 16% and the reversal spread by 30%, consistent with common-factor contamination contributing to the IC pattern. The spread remains large after stripping, however, confirming that correlation convergence is a genuine but partial explanation, with the remainder arising through idiosyncratic channels examined in Section 5.3.

Table 7: Correlation Convergence — Aggregate and Return-Structure Evidence

Panel A: Average Pairwise Correlation by Regime					
Regime	Mean Corr	t-stat	N Months	Cross-Serial Corr	t-stat
Calm	0.110	15.86***	44	+0.011	0.59
Stress	0.229	10.62***	38	-0.025	-1.55
Crash	0.255	11.66***	14	-0.091	-2.16*
Recovery	0.151	17.54***	53	+0.013	0.75
Crash vs. calm difference: 0.145 (Welch $t=5.66^{***}$). Cross-serial correlation crash vs. calm: -0.102 (Welch $t=-2.25^*$).					
Panel B: Horse-Race Regression — Momentum IC on p_crash and Average Pairwise Correlation					
	M1	M2	M3		
p_crash (lag 1)	-0.0136 ($t=-1.42$)	-0.0071 ($t=-0.61$)			
Avg corr (lag 1)		-0.0225 ($t=-2.27^{**}$)	-0.0244 ($t=-3.09^{***}$)		
R ²	0.0221	0.0562	0.0686		

Notes: Panel A uses the Elton-Gruber (1973) equal-weighted portfolio variance decomposition to estimate average pairwise correlation. Cross-serial correlation is the Pearson correlation between month- t and month- $t+1$ cross-sections of stock returns. Panel B reports OLS with Newey-West standard errors (lag=3). The inclusion of average pairwise correlation (M2, M3) substantially reduces the magnitude and significance of the p_crash coefficient relative to M1, consistent with correlation convergence accounting for much of the regime indicator's predictive power. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

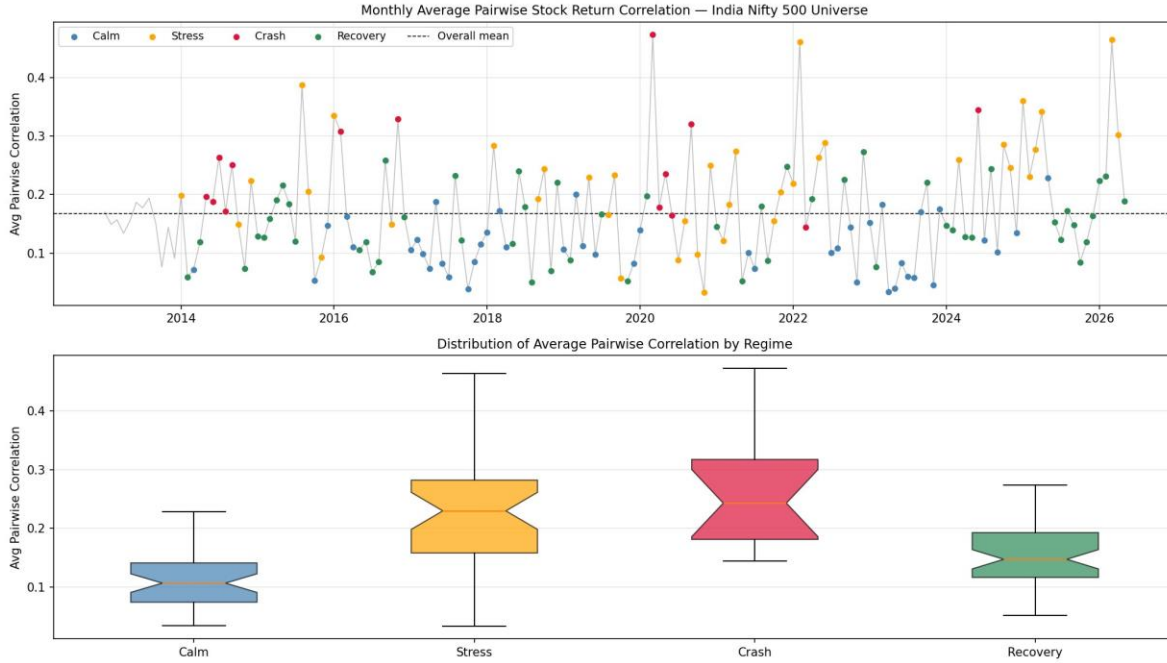


Figure 4: Average Pairwise Stock Return Correlation by Regime, 2013–2026. Top: Monthly time series with regime-coloured dots. Bottom: Box plots by regime. Crash months show mean correlation of 0.25 versus 0.11 in calm months.

5.3. Loser-Specific Overshooting

The complementary mechanism operates through the reversal leg. We propose that crash months create temporary but severe mispricings in recent loser stocks through panic-driven indiscriminate selling — prices are pushed below fair value and recover during the subsequent holding period, generating the positive IC that reversal captures at the monthly level. This interpretation follows a long tradition in market microstructure and behavioural finance linking temporary overreaction to subsequent price correction, dating to De Bondt and Thaler's (1985) overreaction hypothesis and to liquidity-provision models such as Campbell, Grossman, and Wang (1993), in which uninformed selling pressure temporarily depresses prices before liquidity providers restore them toward fundamental value. We present three pieces of evidence consistent with this account.

First, stocks sorted by the reversal signal generate forward returns of approximately zero at all horizons in crash months for the loser quintile — a fade-to-zero rather than continuation pattern — independently confirming Table 2's IC evidence using portfolio returns rather than signal correlations.

Second, we identify every stock-day in which the daily return falls within a tolerance of the stock's large-move threshold and compute market-adjusted CARs over the subsequent one through ten trading days separately for Q1 (momentum losers) and Q5 (momentum winners). In crash months, Q1 stocks that experience a sharp upward price move generate CARs of +2.64% over five days ($t = 3.35$), while Q5 stocks under identical conditions generate only +0.90% ($t = 1.42$) — a difference of +1.74% (Welch $t = 2.38$, $p = 0.017$). In calm months, there is no significant Q1–Q5 difference ($t = -0.41$, $p = 0.68$). The overshooting is both loser-specific and crash-specific, distinguishing it from a generic market-wide bounce.

Third, the bounce is front-loaded: CARs peak at three to five days and remain stable through ten days rather than reversing, consistent with temporary dislocation correcting rather than a fundamental rerating that would be expected to persist or continue growing.

Together, these three pieces of evidence — the fade-to-zero pattern at the portfolio level, the loser-specific CAR asymmetry at the event level, and the front-loaded decay profile of the bounce — paint a consistent picture of panic-driven overshooting in recent loser stocks that corrects within the monthly holding period, and together they account for reversal's positive IC in crash regimes documented in Section 4.1.

Table 8: Microstructure Evidence

Panel A: Momentum IC by Volatility Tercile and Regime					
Vol Tercile	Calm	Stress	Crash	Recovery	Pooled
T1 (Low vol)	+0.069	+0.027	-0.026	+0.054	+0.046
T2 (Mid vol)	+0.073	+0.029	+0.005	+0.011	+0.035
T3 (High vol)	+0.061	-0.023	+0.017	+0.016	+0.029
Panel B: Large Daily Price Moves — Frequency by Volatility Tercile					
Vol Tercile	Calm rate	Crash rate	Crash/Calm ratio	z-stat (vs T3)	
T1 (Low vol)	0.02%	0.96%	45.8×	19.6***	
T2 (Mid vol)	0.22%	1.67%	7.7×		
T3 (High vol)	1.11%	2.77%	2.5×	—	
Panel C: Q1 vs Q5 Post-Bounce Cumulative Abnormal Returns (5% band, crash months)					
Portfolio	1-day CAR	3-day CAR	5-day CAR	10-day CAR	N events
Q1 (momentum losers)	+1.28%***	+2.66%***	+2.64%***	+2.72%*	301
Q5 (momentum winners)	+0.45%*	+0.50%	+0.90%	+1.96%*	318
Q1 – Q5 difference	+0.83%	+2.16%**	+1.74%**	+0.76%	—

Notes: Panel A reports IC computed separately within vol tercile $\times$ regime cells. Panel B reports the fraction of stock-days with return within $\pm 0.3pp$ of the $\pm 5\%$ threshold; the 45.8 $\times$ ratio for low-volatility stocks reflects a rise from 0.02% to 0.96% of trading days — a large relative change from a very low base, paired here with the absolute percentages to avoid overstating the magnitude. Panel C reports market-adjusted CARs following upper-threshold hit events in crash-regime months. CARs = cumulative abnormal returns (market-adjusted). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.4. Within-Universe Volatility Heterogeneity

A novel and somewhat counterintuitive finding connects the two surviving channels at the within-universe level: in crash months, the best momentum predictability lies in the most volatile segment of the universe, not the least. When stocks are sorted into volatility terciles each month and momentum IC is computed separately within each group, the IC gradient across terciles is essentially flat in calm months — low-volatility stocks have IC of +0.069, mid-volatility +0.073,

high-volatility $+0.061$ — but reverses completely in crash months, where low-volatility stocks fall to $IC = -0.026$, mid-volatility falls to near zero ($+0.005$), and high-volatility stocks retain a marginally positive IC ($+0.017$).

This reversal is consistent with differential price-discovery disruption by volatility segment. Low-volatility stocks in calm months experience large daily price moves on approximately 0.02% of trading days; in crash months, this rate rises to 0.96% of days, a 45.8-fold relative increase from a very low base. High-volatility stocks, by contrast, go from 1.1% to 2.8% of days, a 2.5-fold increase from a substantially higher starting point. We report both the relative and absolute figures together throughout, since the multiplier alone can overstate the magnitude of change for a stock category that rarely experiences large moves to begin with. The return process of low-volatility stocks nonetheless undergoes a genuine and economically meaningful transformation in crash months — from stable and fundamentals-driven to comparatively volatile — and their past returns cease to carry useful information about future fundamentals precisely when their price behaviour begins to resemble that of higher-volatility names.

We interpret these findings with appropriate caution. We document that the timing of momentum's IC breakdown coincides with a substantial relative increase in large daily price moves for low-volatility stocks, and that this differential increase is consistent with India's tiered price-band system amplifying price discontinuity in stressed periods. We do not claim causal identification of this channel. A plausible alternative explanation is that realised volatility is itself an imperfect proxy for an omitted liquidity characteristic — low-volatility stocks may also tend to have lower free-float or thinner analyst coverage, in which case volatility classification could be capturing a liquidity effect rather than driving the price-discovery disruption directly.

Distinguishing between these explanations would require stock-level price-band classification and liquidity data linked to individual return outcomes, which we leave for future work.

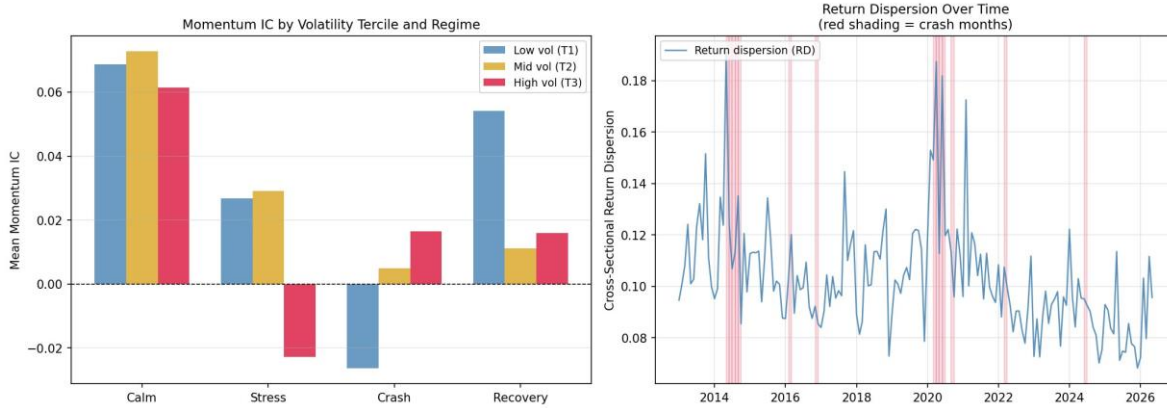


Figure 5: Momentum IC by Volatility Tercile and Regime (left panel) and Return Dispersion Over Time with Crash-Month Shading (right panel). The IC gradient reverses in crash months: low-vol stocks become the worst momentum performers.

6. Strategy Evidence

6.1. Strategy Design

We translate the IC evidence into investable strategies to assess the economic significance of the regime-dependent complementarity documented above. Three strategies are presented, each motivated by a specific prior finding and designed before examining strategy-level returns. S6 (Long-Only Regime Filter) holds the top momentum quintile in calm and recovery months, the top reversal quintile in crash months, and a cash position in stress months — the latter choice reflecting that stress is the regime with the least directional clarity in Table 2, where momentum IC is only marginally positive ($t = 0.32$) and reversal IC, while significant, is smaller in magnitude than in crash, making a conservative cash position the natural choice rather than committing capital to either signal. S7 (Continuous Overlay) weights the momentum quintile by $(1 - p_{\text{crash}})$ and the

reversal quintile by p_crash at each month-end, implementing the continuous relationship from Table 5 directly. S8 (Discrete Overlay) blends full momentum in calm and recovery with a 50/50 momentum-and-equal-weight split in stress and a 50/50 reversal-and-equal-weight split in crash, a practitioner-friendly discretisation of S7.

All three strategies are long-only, motivated by two practical considerations specific to the Indian institutional context: short-selling infrastructure is costly and constrained for Nifty 500 stocks, and dollar-neutral long-short constructions forgo the equity risk premium in a sustained bull market. We confirm this by also reporting S1 (static long-short momentum) as the null hypothesis benchmark; it loses 6.5% annualised over the evaluation period, illustrating the cost of the short leg when market conditions do not support cross-sectional momentum on its own.

All strategies are evaluated out-of-sample from January 2016 after the HMM's 36-month burn-in. Signal formation uses only month- t information to determine positions held in month $t+1$, and the HMM uses filtered posteriors throughout. Transaction costs are applied at 50, 100, and 150 basis points per one-way trade, with turnover measured as the fraction of portfolio weight changed each month.

6.2. Results

Table 9, Panel A, reports net performance at 100 basis points. The long-only strategies deliver strong results: S6 generates 17.4% annualised return with a Sharpe ratio of 0.48, S7 generates 16.8% with Sharpe 0.46, and S8 generates 16.6% with Sharpe 0.47. Panel B shows all three survive 150 basis points with Sharpe ratios of approximately 0.29. Panel C reports monthly alpha relative to the Nifty 500 cap-weight index: S6 delivers annualised alpha of 21.0% ($t = 2.09$),

S7 delivers 19.2% ($t = 2.48$), and S8 delivers 18.1% ($t = 2.44$). The near-zero market betas of S7 and S8 (-0.056 and -0.006) confirm that this alpha is not a disguised leveraged market position.

Panel D shows the regime breakdown of monthly net returns, which makes the source of the strategies' advantage transparent: it is concentrated in crash months. S6 generates +7.2% per crash month, S7 +7.2%, and S8 +6.8%, driven directly by the reversal rotation that the IC evidence and mechanism analysis predicted. Calm and recovery month returns are also positive (+1.5% to +1.6% per month), reflecting the long-only momentum position's participation in the Indian equity bull market of 2016–2026.

The relevant benchmark for evaluating the regime signal's contribution is not a passive investment alternative but rather a static cross-sectional factor strategy that ignores regime information. Against this benchmark — S1, the natural comparison for a cross-sectional factor investor — the regime-conditioned strategies add 17–21 percentage points of annualised alpha ($t = 2.1$ to 2.5). A fund manager who already intends to trade the cross-section of Indian equities benefits materially from incorporating the regime signal; the relevant question is whether conditioning on the regime improves their active strategy, and the evidence indicates clearly that it does.

For completeness, we also benchmark against a constructed equal-weight portfolio of the same 786-stock universe, which earned 20.5% annualised with a Sharpe ratio of 0.70 over the same period. The strategies do not generate statistically significant alpha relative to this benchmark ($t = 1.48$ for S7), a result consistent with the well-documented equal-weight premium in Indian equities — a market-structural feature affecting all active strategies in this universe rather than a deficiency specific to the regime signal. The defence of the strategies' value is more specific than

the general level comparison, however: their advantage over equal-weight, even though not significant in aggregate, is concentrated precisely in crash months, where the regime-conditioned strategies generate +6.8% to +7.2% per month (Panel D) — exactly when the regime signal is designed to add value, and a pattern an unconditional equal-weight buy-and-hold portfolio cannot replicate by construction. The NSE 500 Equal Weight Index was, in any case, not available as an investable vehicle for most of the evaluation period, having been launched only in May 2024.

Table 9: Strategy Performance (Out-of-Sample, January 2016 – May 2026)

Panel A: Net Performance at 100bps TC (OOS Jan 2016 – May 2026)							
Strategy	Ann Ret	Ann Vol	Sharpe	Max DD	Alpha (ann)	t-Alpha	Avg TO
S1: Static Momentum	-6.5%	17.2%	-0.73	-56.4%	-7.6%	-1.42	91%
S6: Long-Only Regime	17.4%	23.5%	0.48	-35.6%	21.0%	2.09**	66%
S7: Continuous Overlay	16.8%	23.5%	0.46	-37.5%	19.2%	2.48**	63%
S8: Discrete Overlay	16.6%	22.6%	0.47	-36.0%	18.1%	2.44**	59%
Panel B: Sharpe Ratio Sensitivity to Transaction Costs							
Strategy	50bps	100bps	150bps				
S1: Static Momentum	-0.42	-0.73	-1.02				
S6: Long-Only Regime	+0.68	+0.48	+0.29				
S7: Continuous Overlay	+0.64	+0.46	+0.28				
S8: Discrete Overlay	+0.66	+0.47	+0.29				
Panel C: Monthly Net Return by Regime (100bps TC, %)							
Strategy	Calm	Stress	Crash	Recovery	All		
S1: Static Momentum	+0.61%	-0.84%	-4.63%	-0.24%	-0.43%		
S6: Long-Only Regime	+1.55%	cash	+7.20%	+0.43%	+1.58%		
S7: Continuous Overlay	+1.60%	+1.39%	+7.17%	+0.41%	+1.54%		
S8: Discrete Overlay	+1.52%	+1.47%	+6.82%	+0.40%	+1.51%		

Notes: All strategies evaluated out-of-sample from January 2016 (after 36-month HMM burn-in). Alpha estimated vs. the Nifty 500 cap-weight index using OLS with Newey-West standard errors (lag=3). TO = average monthly portfolio turnover. Max DD = maximum drawdown. S6 holds cash in stress months, reflecting the absence of a clear directional signal for either momentum or reversal in that regime (Table 2: momentum IC $t=0.32$, reversal IC marginally significant). ** $p < 0.05$.

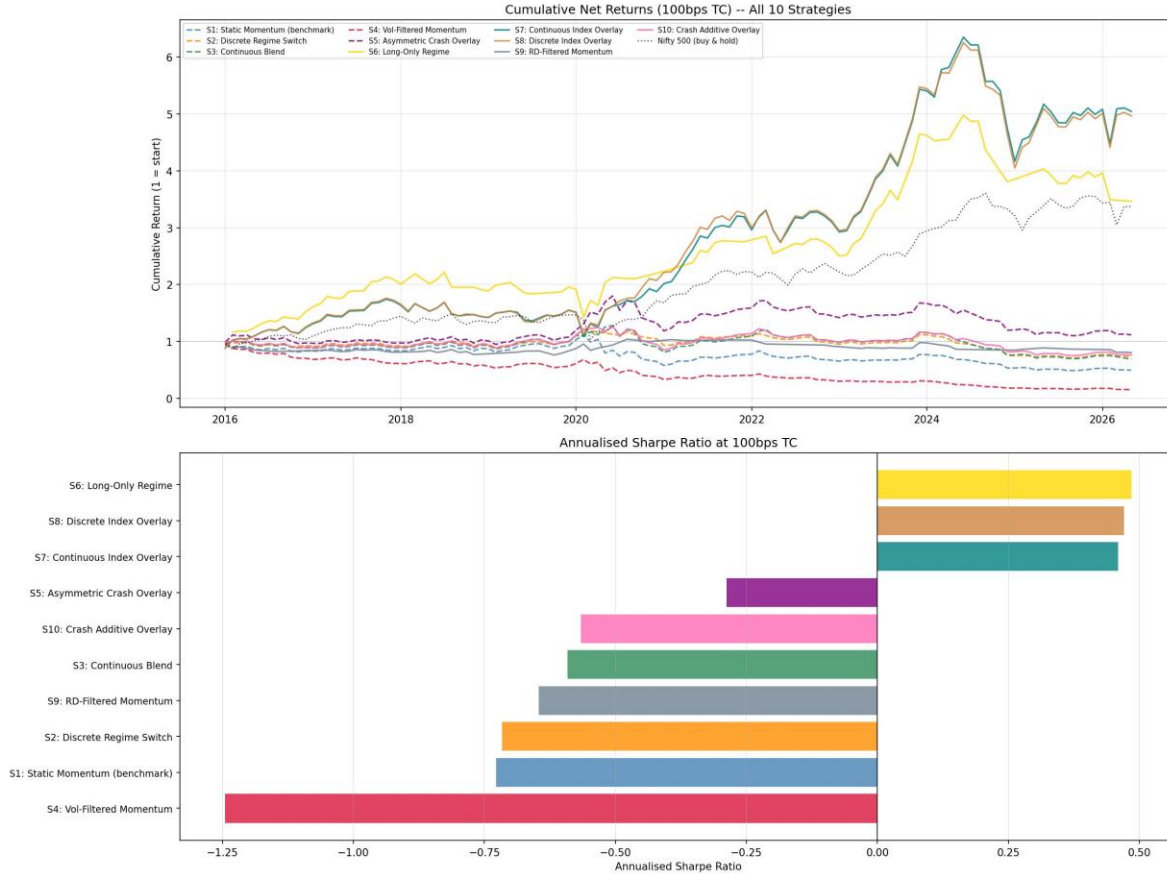


Figure 6: Cumulative Net Returns for All Strategies (100bps TC), January 2016 – May 2026. Solid lines: long-only strategies (S6, S7, S8). Dashed lines: long-short strategies (S1–S5). Dotted: Nifty 500 buy-and-hold. The long-only regime strategies spike in 2020 (crash) and maintain their advantage.

6.3. Validation of the Continuous Relationship

The continuous blending strategy S3 — a long-short version of S7 — provides a direct out-of-sample test of the Table 5 continuous relationship. We regress S3's excess return over S1 on the lagged HMM crash probability:

$$S3_return - S1_return = \alpha + \beta \cdot p_crash + \varepsilon$$

Although S3 does not produce positive absolute returns on its own (-3.5% annualised, consistent with the weak long-short momentum premium in this period), the regression yields $\beta = 0.057$ ($t = 2.13$, $R^2 = 0.13$): S3 outperforms S1 by 1.33% per month in the highest p_crash quartile and is essentially neutral in the lowest quartile. This monotone relationship between crash probability and relative strategy performance is the out-of-sample portfolio analogue of the Table 5 regression, and confirms that the continuous blending adds value precisely when and where the IC evidence predicts it should.

7. Conclusion

This paper establishes that momentum and reversal signals operate as regime-dependent complements in Indian equities: momentum dominates in calm market states, reversal dominates in crashes, and the spread between their information coefficients shifts continuously as the HMM-estimated crash probability rises. This continuous relationship, together with its real-time implementability, is the paper's primary contribution, confirmed out-of-sample in Fama-MacBeth regressions ($t = 3.03$ on the calm–crash coefficient difference) and in portfolio tests where the continuous blending strategy's outperformance over static momentum increases monotonically with crash probability.

We evaluate six candidate explanations for this pattern and find evidence most consistent with two channels. Pairwise correlation convergence is the strongest supported channel among those examined: correlations rise from 0.11 to 0.25 in crash regimes, largely accounting for the regime indicator's predictive power in a horse-race regression and operating at the return-structure level through a sign reversal of cross-sectional serial correlation from $+0.011$ in calm to -0.091 in

crash. A second, complementary channel — loser-specific overshooting — is documented at the daily frequency: recent loser stocks generate +2.64% market-adjusted CARs over five days after sharp upward price moves in crash months, significantly larger than the +0.90% observed for winners. A novel within-universe finding shows that low-volatility stocks experience a substantially larger relative increase in large daily price moves during crash months than high-volatility stocks, consistent with momentum's IC breakdown concentrating in the lower-volatility segment. Four alternative mechanisms — foreign institutional flows, option-like loser volatility, return dispersion compression, and beta asymmetry — find little support, each strengthening confidence in the channels that do.

The strategy evidence establishes economic significance: regime-conditioned long-only strategies deliver Sharpe ratios of 0.46–0.48 after 100 basis points in transaction costs, generating statistically significant alpha of 18–21% annualised relative to the cap-weight Nifty 500 index, concentrated in crash months exactly as the mechanism evidence predicts. The strategies do not significantly outperform a constructed equal-weight benchmark in aggregate, a result reflecting the well-documented equal-weight premium in Indian equities rather than a limitation of the regime signal, though their advantage over equal-weight is itself concentrated in crash months in the same pattern as their advantage over the cap-weight index.

We document a robust empirical pattern and provide evidence consistent with correlation convergence and crash-period price-discovery disruption as the primary economic channels. We do not claim definitive causal identification of the full mechanism linking these channels, and several limitations bound the scope of these conclusions. The analysis covers the Nifty 500 large-cap universe from 2013 to 2026; results may not generalise to smaller-capitalisation segments, other emerging markets, or shorter holding periods. The crash sample is necessarily limited ($n =$

14 months in the full sample, $n = 9$ in the post-2015 subsample), though the out-of-sample Fama-MacBeth validation ($t = 3.03$) and the post-2015 robustness check substantially mitigate concerns about statistical power. Causal identification of the within-universe volatility heterogeneity channel would require stock-level price-band classification data linked to individual return outcomes across multiple market cycles, which we leave for future research, alongside extension to other emerging markets with similar price-band microstructure and higher-frequency implementation using daily signal updating within the HMM framework.

Declarations

CRedit Authorship Contribution Statement: Mrigank Shekhar Sahi: Conceptualisation, Methodology, Software, Formal analysis, Data curation, Writing – original draft, Writing – review and editing, Visualisation.

Data Availability Statement: The data used in this study are derived from publicly available sources. Daily OHLCV data are available from Yahoo Finance (finance.yahoo.com). NSE Nifty 500 constituent history is available from the National Stock Exchange of India (nseindia.com) and Bloomberg. Processed datasets and analysis code are available from the corresponding author on reasonable request.

Conflict of Interest: The author declares no conflict of interest.

References

- Amihud, Y. (2002). Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets*, 5(1), 31–56.
- Ang, A., & Chen, J. (2002). Asymmetric correlations of equity portfolios. *Journal of Financial Economics*, 63(3), 443–494.
- Ansari, V. A., & Khan, S. (2012). Momentum anomaly: Evidence from India. *Managerial Finance*, 38(2), 206–223.
- Asness, C. S., Frazzini, A., & Pedersen, L. H. (2013). Quality minus junk. *Review of Accounting Studies*, 24(1), 34–112.
- Barroso, P., & Santa-Clara, P. (2015). Momentum has its moments. *Journal of Financial Economics*, 116(1), 111–120.
- Campbell, J. Y., Grossman, S. J., & Wang, J. (1993). Trading volume and serial correlation in stock returns. *Quarterly Journal of Economics*, 108(4), 905–939.
- Daniel, K., & Moskowitz, T. J. (2016). Momentum crashes. *Journal of Financial Economics*, 122(2), 221–247.
- De Bondt, W. F. M., & Thaler, R. (1985). Does the stock market overreact? *Journal of Finance*, 40(3), 793–805.
- Dimson, E. (1979). Risk measurement when shares are subject to infrequent trading. *Journal of Financial Economics*, 7(2), 197–226.
- Elton, E. J., & Gruber, M. J. (1973). Estimating the dependence structure of share prices. *Journal of Finance*, 28(5), 1203–1232.
- Fama, E. F., & MacBeth, J. D. (1973). Risk, return, and equilibrium: Empirical tests. *Journal of Political Economy*, 81(3), 607–636.
- Guidolin, M., & Timmermann, A. (2007). Asset allocation under multivariate regime switching. *Journal of Economic Dynamics and Control*, 31(11), 3503–3544.
- Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357–384.
- Jegadeesh, N. (1990). Evidence of predictable behavior of security returns. *Journal of Finance*, 45(3), 881–898.
- Jegadeesh, N., & Titman, S. (1993). Returns to buying winners and selling losers: Implications for stock market efficiency. *Journal of Finance*, 48(1), 65–91.
- Lehmann, B. N. (1990). Fads, martingales, and market efficiency. *Quarterly Journal of Economics*, 105(1), 1–28.
- Lewellen, J. (2002). Momentum and autocorrelation in stock returns. *Review of Financial Studies*, 15(2), 533–563.

- Longin, F., & Solnik, B. (2001). Extreme correlation of international equity markets. *Journal of Finance*, 56(2), 649–676.
- Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Patel, S. A. (2012). Momentum effect in Indian stock market. *Global Finance Journal*, 23(3), 147–161.
- Rouwenhorst, K. G. (1998). International momentum strategies. *Journal of Finance*, 53(1), 267–284.
- Sehgal, S., & Jain, S. (2011). Short-term momentum patterns in stock and sectoral returns: Evidence from India. *Journal of Advances in Management Research*, 8(1), 99–122.
- Stivers, C., & Sun, L. (2010). Cross-sectional return dispersion and the equity premium. *Journal of Financial and Quantitative Analysis*, 45(3), 563–594.